PTF ALL

Equip Cooling Overboard Exhaust Valve Does Not Open When Commanded - Fault Isolation

CORRECTIVE ACTION SUMMARY

Replace Overboard Exhaust Valve Open Enable Relay [R650]
Replace Overboard Exhaust Valve [V157]
Replace Overboard Exhaust Valve Open Control Relay [R649]
Replace Forward Fire Alarm Latch Relay [R945]
Replace Recirculate Fan Switch Bypass Relay [R644]
Replace Smoke Control Relay [R648]
Replace Cabin Pressure Control Module [P5-6]
Repair Wiring D11878-D11880
Repair Wiring D11878-Circuit Breaker C1444
Repair Wiring D11874-D11878
Repair Wiring D11866-D12225
Repair Wiring D11874-D11866
Repair Wiring D11874-Circuit Breaker C1444
Repair Wiring Circuit Breaker C1445-D11866
Repair Wiring D12225-D11878

A. Description

(1) This task is for this observed fault:

(a) The equipment cooling overboard exhaust valve does not open when commanded.

B. Initial Evaluation

(1) Do this task:

- Smoke Clearance Mode - Operational Test, AMM TASK 21-27-00-700-807.

(a) If the system does not operate as specified in the AMM task, then do the Fault Isolation Procedure below.

(b) If the system operates as specified in the AMM task, then there was an intermittent fault.

C. Fault Isolation Procedure

(1) Replace the overboard exhaust valve open enable relay, R650.

(a) Do the AMM test that you did at the start of this task.

(b) If the system operates as specified in the AMM task, then you corrected the fault.

(c) If the system does not operate as specified in the AMM task, then continue.

(2) Replace the overboard exhaust valve, V157. These are the tasks:

- Overboard Exhaust Valve Removal, AMM TASK 21-43-01-000-801,
- Overboard Exhaust Valve Installation, AMM TASK 21-43-01-400-801.

(a) Do the AMM test that you did at the start of this task.

(b) If the system operates as specified in the AMM task, then you corrected the fault.

(c) If the system does not operate as specified in the AMM task, then continue.

(3) Replace the overboard exhaust valve open control relay, R649.

(a) Do the AMM test that you did at the start of this task.

(b) If the system operates as specified in the AMM task, then you corrected the fault.

(c) If the system does not operate as specified in the AMM task, then continue.

(4) Replace the forward fire alarm latch relay, R945.

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(a) Do the AMM test that you did at the start of this task.

(b) If the system operates as specified in the AMM task, then you corrected the fault.

(c) If the system does not operate as specified in the AMM task, then continue.

(5) Replace the recirculate fan switch bypass relay, R644.

(a) Do the AMM test that you did at the start of this task.

(b) If the system operates as specified in the AMM task, then you corrected the fault.

(c) If the system does not operate as specified in the AMM task, then continue.

(6) Replace the smoke control relay, R648.

(a) Do the AMM test that you did at the start of this task.

(b) If the system operates as specified in the AMM task, then you corrected the fault.

(c) If the system does not operate as specified in the AMM task, then continue.

(7) Replace the cabin pressure control module, P5-6. These are the tasks:

- Cabin Pressure Control Module Removal, AMM TASK 21-31-02-000-801,
- Cabin Pressure Control Module Installation, AMM TASK 21-31-02-400-801.

(a) Do the AMM test that you did at the start of this task.

(b) If the system operates as specified in the AMM task, then you corrected the fault.

(c) If the system does not operate as specified in the AMM task, then continue.

(8) Do this check of the wiring:

(a) Disconnect connector D11878 from the overboard exhaust valve open enable relay, R650.

(b) Disconnect connector D11880 from the overboard exhaust valve, V157.

(c) Do a wiring check between this pin of connector D11878 at the overboard exhaust valve open enable relay and this pin of connector D11880 at the overboard exhaust valve (WDM 21-43-21):

D11878		**D11880**
pin B1	.....	pin 2

(d) Do a wiring check to make sure that pin X2 of D11878 has continuity to ground.

(e) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect connector D11878 to the overboard exhaust valve open enable relay, R650.

3) Re-connect connector D11880 to the overboard exhaust valve, V157.

4) Do the AMM test that you did at the start of this task.

5) If the system operates as specified in the AMM task, then you corrected the fault.

(f) If you do not find a problem with the wiring, then continue.

(g) Re-connect connector D11880 to the overboard exhaust valve, V157.

(9) Do this check of the wiring:

(a) Open the A/C OVERBOARD EXHAUST VALVE CONT circuit breaker, C1444 [P6-4, D1].

(b) Do a wiring check between this pin of connector D11878 at the overboard exhaust valve open enable relay and this terminal of circuit breaker C1444 (WDM 21-43-21):

D11878		**C1444**
pin B2	.....	term =L

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect connector D11878 to the overboard exhaust valve open enable relay, R650.

3) Close the A/C OVERBOARD EXHAUST VALVE CONT circuit breaker, C1444 [P6-4, D1].

4) Do the AMM test that you did at the start of this task.

5) If the system operates as specified in the AMM task, then you corrected the fault.

(d) If you do not find a problem with the wiring, then continue.

(e) Close the A/C OVERBOARD EXHAUST VALVE CONT circuit breaker, C1444 [P6-4, D1].

(10) Do this check of the wiring:

(a) Disconnect connector D11874 from the smoke control relay, R648.

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(b) Do a wiring check between this pin of connector D11874 at the smoke control relay and this pin of connector D11878 at the overboard exhaust valve open enable relay (WDM 21-43-21):

D11874		D11878
pin B1	.....	pin A2

(c) If you find a problem with the wiring, then do these steps:
1) Repair the wiring.
2) Re-connect connector D11874 to the smoke control relay, R648.
3) Re-connect connector D11878 to the overboard exhaust valve open enable relay, R650.
4) Do the AMM test that you did at the start of this task.
5) If the system operates as specified in the AMM task, then you corrected the fault.

(d) If you do not find a problem with the wiring, then continue.

(e) Re-connect connector D11874 to the smoke control relay, R648.

(f) Re-connect connector D11878 to the overboard exhaust valve open enable relay, R650.

(11) Do this check of the wiring:

(a) Disconnect connector D11866 from the recirculate fan switch bypass relay, R644.

(b) Disconnect connector D12225 from the forward fire alarm latch relay, R945.

(c) Do a wiring check between these pins of connector D11866 at the recirculate fan switch bypass relay and these pins of connector D12225 at the forward fire alarm latch relay (WDM 21-43-21):

D11866		D12225
pin 4	.....	pin 7
pin 14	.....	pin 11

(d) If you find a problem with the wiring, then do these steps:
1) Repair the wiring.
2) Re-connect connector D11866 to the recirculate fan switch bypass relay, R644.
3) Re-connect connector D12225 to the forward fire alarm latch relay, R945.
4) Do the AMM test that you did at the start of this task.
5) If the system operates as specified in the AMM task, then you corrected the fault.

(e) If you do not find a problem with the wiring, then continue.

(f) Re-connect connector D12225 to the forward fire alarm latch relay, R945.

(12) Do this check of the wiring:

(a) Disconnect connector D11874 from the smoke control relay, R648.

(b) Do a wiring check between these pins of connector D11874 at the smoke control relay and these pins of connector D11866 at the recirculate fan switch bypass relay (WDM 21-43-21):

D11874		D11866
pin B1	.....	pin 5
pin C1	.....	pin 7

(c) If you find a problem with the wiring, then do these steps:
1) Repair the wiring.
2) Re-connect connector D11874 to the smoke control relay, R648.
3) Re-connect connector D11866 to the recirculate fan switch bypass relay, R644.
4) Do the AMM test that you did at the start of this task.
5) If the system operates as specified in the AMM task, then you corrected the fault.

(d) If you do not find a problem with the wiring, then continue.

(e) Re-connect connector D11866 to the recirculate fan switch bypass relay, R644.

(13) Do this check of the wiring:

(a) Open the A/C OVERBOARD EXHAUST VALVE CONT circuit breaker, C1444 [P6-4, D1].

(b) Do a wiring check between these pins of connector D11874 at the smoke control relay and this terminal of circuit breaker C1444 (WDM 21-43-21):

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D11874		C1444
pin B2	.....	term =L
pin C2	.....	term =L

(c) If you find a problem with the wiring, then do these steps:
1) Repair the wiring.
2) Re-connect connector D11874 to the smoke control relay, R648.
3) Close the A/C OVERBOARD EXHAUST VALVE CONT circuit breaker, C1444 [P6-4, D1].
4) Do the AMM test that you did at the start of this task.
5) If the system operates as specified in the AMM task, then you corrected the fault.

(d) If you do not find a problem with the wiring, then continue.

(e) Re-connect connector D11874 to the smoke control relay, R648.

(f) Close the A/C OVERBOARD EXHAUST VALVE CONT circuit breaker, C1444 [P6-4, D1].

(14) Do this check of the wiring:

(a) Open the A/C OVERBOARD EXH VALVE RECONFIG CONT circuit breaker, C1445 [P6-4, D2].

(b) Disconnect connector D11866 from the recirculate fan switch bypass relay, R644.

(c) Do a wiring check between this terminal of circuit breaker C1445 and this pin of connector D11866 at the recirculate fan switch bypass relay (WDM 21-43-21):

C1445		D11866
term =L	.....	pin 6

(d) Do a wiring check to make sure that pin 12 of D11866 has continuity to ground.

(e) If you find a problem with the wiring, then do these steps:
1) Repair the wiring.
2) Close the A/C OVERBOARD EXH VALVE RECONFIG CONT circuit breaker, C1445 [P6-4, D2].
3) Re-connect connector D11866 to the recirculate fan switch bypass relay, R644.
4) Do the AMM test that you did at the start of this task.
5) If the system operates as specified in the AMM task, then you corrected the fault.

(f) If you do not find a problem with the wiring, then continue.

(g) Close the A/C OVERBOARD EXH VALVE RECONFIG CONT circuit breaker, C1445 [P6-4, D2].

(h) Re-connect connector D11866 to the recirculate fan switch bypass relay, R644.

(15) Do this check of the wiring:

(a) Disconnect connector D12225 from the forward fire alarm latch relay, R945.

(b) Disconnect connector D11878 from the overboard exhaust valve open enable relay, R650.

(c) Do a wiring check between this pin of connector D12225 at the forward fire alarm latch relay and this pin of connector D11878 at the overboard exhaust valve open enable relay (WDM 21-43-21):

D12225		D11878
pin 8	.....	pin X1

(d) If you find a problem with the wiring, then do these steps:
1) Repair the wiring.
2) Re-connect connector D12225 to the forward fire alarm latch relay, R945.
3) Re-connect connector D11878 to the overboard exhaust valve open enable relay, R650.
4) Do the AMM test that you did at the start of this task.
5) If the system operates as specified in the AMM task, then you corrected the fault.

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